ENGINEERING PORTFOLIO

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Mechanical Engineering · Liquid Propulsion & Fluid Systems

> I design, build, and test liquid rocket propulsion hardware. My work has moved from single components, to owning a subsystem that flew, to architecting a full cryogenic fluid system from a blank sheet.

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This portfolio is organized as a timeline. Each project records the problem, the constraints, the reasoning behind the design decisions, and how the result was validated. The through line is a progression from component design, to subsystem ownership on a vehicle that flew, to full-system architecture on a cryogenic stand.

FS-03

Project Clementine

LOX / Ethanol Liquid Bipropellant Rocket

ROLE	ORGANIZATION	PERIOD
Vice President; previously Lead Fluid Systems Engineer	Highlander Space Program, UC Riverside	2026 - Present

The problem

Our previous vehicle, Project Poseidon, used nitrous oxide, which is self-pressurizing: the propellant supplies its own tank pressure through its vapor pressure, so there is no separate pressurization system to design. Moving to liquid oxygen removes that convenience. LOX has to be pushed, which means the whole pressurization architecture becomes something I had to design rather than inherit.

I owned the P&ID for that system: selecting the pneumatics, valves, and regulators, setting the pressure schedule, defining the interfaces between the pressurization, propellant, and ignition subsystems, and installing the result on our fluids test wall.

Constraints

- **Single pressurant source.** One GN2 bottle had to serve both the pneumatic actuation and the propellant tanks. Budget did not allow separate supplies.
- **Cryogenic propellant.** LOX introduces material compatibility, thermal, and trapped volume hazards that a room-temperature system does not have.
- **Student build.** Commercial off-the-shelf components, limited machining, and a team where members rotate in and need to operate the system safely without deep context.
- **Every test is a hazardous operation.** The system had to be safe to operate and legible to a new operator, not just correct on paper.

Design decisions

Holding tank pressure constant as the bottle drains

A single GN2 bottle blows down as gas is consumed. Treating the bottle as ideal gas,

$$\frac{p_1 V}{T_1} = \frac{p_2 V}{T_2}$$

supply pressure falls continuously through a test. A single spring regulator tracks that decay: spring regulators exhibit **droop**, where outlet pressure falls as inlet pressure drops and as flow increases, because the spring force is fixed while the sensing area sees changing conditions.

Inconsistent tank pressure means inconsistent chamber pressure, which means the engine is not being tested at the condition it was designed for.

My solution was a **two-stage regulation scheme**: a spring regulator sets the reference pressure in the dome of a dome-loaded regulator. The dome regulator references that near-constant dome pressure across a large diaphragm area rather than a spring, so its outlet stays essentially flat as the bottle decays and as flow varies. The spring regulator only has to hold a low-flow reference, which is the job it does well.

Sizing the pressurant lines

During checkout I measured a pressure drop across the GN2 lines that was larger than the budget allowed. Rather than raise the supply pressure to compensate, I traced it to undersized tubing and corrected the geometry.

The reasoning is the Darcy-Weisbach relation for major losses,

$$\Delta p = f\,\frac{L}{D}\,\frac{\rho v^2}{2}$$

with the flow velocity set by continuity,

$$v = \frac{4Q}{\pi D^2}$$

Substituting velocity into the pressure drop shows why line diameter dominates:

$$\Delta p \;\propto\; \frac{1}{D}\cdot\frac{1}{D^4} \;=\; \frac{1}{D^5}$$

At fixed volumetric flow, pressure drop scales with the **inverse fifth power** of internal diameter. Going up a single tube size is worth far more than any amount of tuning elsewhere in the run, and it fixes the problem at its source instead of masking it with higher supply pressure. Upsizing the GN2 lines brought the drop back inside budget.

Fittings, valves, and bends were then added as minor losses,

$$\Delta p_{\text{minor}} = K\,\frac{\rho v^2}{2}, \qquad \Delta p_{\text{total}} = \sum \Delta p_{\text{major}} + \sum \Delta p_{\text{minor}}$$

with the friction factor from the Reynolds number,

$$Re = \frac{\rho v D}{\mu}$$

Relief valves on every isolable section

Any volume that can be closed off at both ends by ball valves can trap fluid. With a cryogenic system this is not a theoretical concern: trapped liquid oxygen warming toward ambient expands by roughly a factor of 860 from liquid to gas, and in a rigid closed volume that becomes a pressure rise that will find the weakest component.

I walked the P&ID section by section, identified every volume that could be isolated between two valves, and placed a relief valve on each one. The rule I designed to was that no operator sequence, correct or incorrect, should be able to create a sealed volume with no path to relieve.

Instrumentation at every node

Each measurement point carries **both** a mechanical pressure gauge and a pressure transducer. This is deliberate redundancy with two different failure modes:

- The **transducer** feeds the DAQ, giving a continuous recorded trace for post-test analysis and live monitoring.
- The **gauge** is independent of power and software. An operator can confirm the state of the system by eye during a walkdown, and a disagreement between gauge and transducer immediately separates an instrumentation fault from a real fluid anomaly.

That distinction matters during a no-go: knowing whether the sensor is lying or the system is genuinely off-nominal determines whether you safe the stand or keep working.

Igniter propellant split

The tanks divert a small portion of fuel and oxidizer to a torch igniter that lights the main combustion chamber. Sizing that bleed is an orifice problem,

$$\dot{m} = C_d\, A \sqrt{2\rho\,\Delta p}$$

The split has to deliver a reliable igniter flow without meaningfully disturbing the pressure or flow delivered to the main injector.

What I built

A complete pressurization, propellant, and ignition fluid system: GN2 supply feeding both pneumatic actuation and tank pressurization through two-stage regulation, run tanks for fuel and oxidizer, poppet pneumatic valve actuation, a torch igniter feed, relief protection on every isolable section, and instrumentation at each node reporting to the DAQ. The system is installed on our fluids test wall and is the stand the team now tests on.

Validation

1. **Component bench testing** before integration, to confirm each regulator, valve, and transducer behaved as specified rather than as assumed.
2. **System pressure checks** using the installed transducers, comparing measured drops against the values predicted by the sizing model.
3. **Correction and re-test**: the GN2 line pressure drop was identified here, corrected by upsizing, and confirmed against the model.

What I would do differently

I would build the pressure drop budget before selecting tubing rather than confirming it during checkout. The physics was not a surprise; I had the sizing tool that computes it. Running the numbers first would have caught the undersized run on paper and saved a rework cycle on hardware. It is the clearest lesson I have taken from this system: analysis is cheapest before you buy the fittings.

FS-02

Project Poseidon

N_2O / Ethanol Liquid Bipropellant Rocket

ROLE	ORGANIZATION	PERIOD
Lead Fluid Systems Engineer	Highlander Space Program, UC Riverside	2025 - 2026

COMPETITION RESULT
1st place, Most Efficient Liquid Engine

EVENT
2025 FAR-OUT Advanced Propulsion Rocketry Competition

FLIGHT ALTITUDE
7,340 ft

The problem

Poseidon was UC Riverside's first liquid bipropellant rocket. Nothing existed to inherit: no prior feed system, no validated pressure vessel, no established test procedure. As Lead Fluid Systems Engineer I owned the pressure vessel and the integrated fuel and oxidizer feed system, from analysis through fabrication and test.

Constraints

- **First of its kind for the program.** No internal heritage hardware or prior data to design against.
- **Competition deadline.** A fixed date that testing had to be complete before.
- **Student budget and shop access.** Design had to be manufacturable with the machining and assembly capability actually available to us.
- **Safety margin on a pressurized structure** with people standing near it.

Design decisions

Why nitrous oxide simplified the pressurization problem

N_2O is self-pressurizing. At room temperature its vapor pressure sits near 700 to 750 psi, so the propellant supplies its own tank pressure without a separate pressurant system. For a first liquid vehicle this removed an entire subsystem and let the team concentrate on the engine, the structure, and the feed path.

The tradeoff is that vapor pressure is strongly temperature dependent, so tank pressure, and therefore engine performance, varies with ambient conditions on test day. That sensitivity is the reason our next vehicle moved to a separately pressurized architecture, which became Project Clementine.

Sizing the pressure vessel

The vessel was analyzed as a thin-walled cylinder, valid where the radius to thickness ratio is greater than about 10. Under internal pressure the wall carries a hoop stress and an axial stress,

$$\sigma_h = \frac{pr}{t}, \qquad \sigma_a = \frac{pr}{2t}$$

Hoop stress is twice axial stress, so the hoop direction governs and sets the required wall thickness,

$$t \ \geq \ \frac{p_{\text{design}}\, r}{\sigma_{\text{allow}}}, \qquad \sigma_{\text{allow}} = \frac{\sigma_y}{FS}$$

Design pressure was set from the maximum expected operating pressure with margin applied above it, rather than from nominal operating pressure, so that the structure is sized for the worst credible case rather than the intended one. The combined stress state was checked against yield using the von Mises criterion,

$$\sigma_v = \sqrt{\sigma_h^2 - \sigma_h \sigma_a + \sigma_a^2}$$

Material selection traded strength to weight against weldability and oxidizer compatibility, and the geometry was iterated in CAD against what the shop could actually produce and inspect.

Designing the feed path

The feed system had to deliver both propellants to the engine at the correct flow and pressure, sequence safely through fill and fire, and seal reliably. Line and orifice sizing followed the same incompressible relations used across the program,

$$\dot{m} = C_d\, A \sqrt{2\rho\, \Delta p}$$

with major and minor losses accumulated along each run to confirm the engine saw its design inlet condition rather than whatever the plumbing happened to deliver.

What I built

The pressure vessel, co-designed and fabricated: structural analysis, CAD iteration, and coordination with manufacturing until it was real hardware. Then the integrated fuel and oxidizer feed system, assembling and validating the feed lines, valves, and interfaces that connected tanks to engine.

Validation

Testing followed a deliberate ladder, each step qualifying the next:

1. **Leak and pressure checks** on the assembled system, to prove sealing before introducing propellant.
2. **Cold flow testing**, flowing propellant through the complete feed path without combustion. This validates flow rates, sequencing, and valve behavior while the consequences of being wrong are still small.
3. **Static fire**, running the full engine with the vehicle restrained, to confirm the feed system delivered propellant correctly under real operating and thermal conditions.
4. **Flight.**

The vehicle earned first place for Most Efficient Liquid Engine at the 2025 FAR-OUT Advanced Propulsion Rocketry Competition and reached 7,340 feet.

> The pressure vessel that flew was drilled the night before the hot fire using a jig I designed and manufactured on the spot. That story is written up separately in Bulkhead Drill Jig.

What I would do differently

I would instrument more heavily. We proved the system worked, but with more transducers across the feed path we would have had a far richer picture of *how* it worked, and a dataset to design the next vehicle against instead of starting Clementine's numbers from first principles again. The instrumentation density on Clementine is a direct response to this.

SW-01

Fluid System Sizing & Trade Study Tool

Python / MATLAB design automation

ROLE	ORGANIZATION	PERIOD
Author	Highlander Space Program, UC Riverside	2025 - Present

The problem

Every change to a feed system propagates. Change a tube size and the velocity, Reynolds number, friction factor, and pressure drop all move, which changes the tank pressure you need, which changes the regulator setpoint and the structural requirement on the tank. Doing that by hand takes long enough that in practice you stop exploring alternatives and commit to the first workable answer.

I wanted the team to be able to ask "what if this line were larger" and get an answer immediately, so that design decisions came from comparison rather than from whichever configuration someone happened to calculate first.

What it computes

Given propellant properties, target mass flow, and a proposed line and tank geometry, the tool returns total pressure drop, flow rates, and line and tank sizing.

Flow velocity from continuity:

$$v = \frac{4Q}{\pi D^2}$$

Reynolds number to establish the flow regime:

$$Re = \frac{\rho v D}{\mu}$$

Friction factor from the Swamee-Jain explicit approximation, which avoids iterating the implicit Colebrook equation:

$$f = \frac{0.25}{\left[\log_{10}\left(\frac{\varepsilon}{3.7D} + \frac{5.74}{Re^{0.9}}\right)\right]^2}$$

Major losses along each run:

$$\Delta p_{\text{major}} = f\,\frac{L}{D}\,\frac{\rho v^2}{2}$$

Minor losses through valves, elbows, and fittings:

$$\Delta p_{\text{minor}} = K\,\frac{\rho v^2}{2}$$

Summed to a system total, which sets the tank pressure required to deliver the design inlet condition at the injector:

$$p_{\text{tank}} = p_{\text{injector}} + \sum \Delta p$$

Why it mattered

The tool turned line sizing into a trade study rather than a guess. It is also what made the diagnosis on Project Clementine immediate: when the measured GN2 pressure drop came in high, the model said what the drop *should* have been for that geometry, which pointed straight at undersized tubing instead of leaving us to hunt.

What I would do differently

I would build compressible flow handling in from the start. The incompressible relations are correct for the liquid propellant runs, but the high-pressure GN2 side is better described with compressibility and choked flow limits accounted for,

$$\frac{p^*}{p_0} = \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}} \approx 0.528 \quad (\gamma = 1.4)$$

so that the tool covers the pressurant side with the same rigor as the propellant side.

MF-01

Bulkhead Drill Jig

Overnight fixture design under a test deadline

ROLE	ORGANIZATION	PERIOD
Designer and machinist	Highlander Space Program, UC Riverside	2025

The problem

The night before we were due to travel out and hot fire the rocket, the pressure vessel was not prepped. The bulkheads could not be installed because the holes at the ends had not been drilled. Without those holes there was no vessel, and without the vessel there was no test.

Constraints

- **One night.** The team was leaving in the morning.
- **No second chance on the part.** This was the flight pressure vessel. A misplaced hole would scrap a component we could not replace before the test.
- **Repeatability across both ends**, and hole positions accurate enough that the bulkhead fastener pattern would actually line up.

Design decisions

The instinct under time pressure is to mark out the holes by hand and drill. I did not, because hand layout puts the positional accuracy of a flight part in a scribe line and a steady hand, and it has to be repeated independently at both ends. Each layout is a fresh opportunity to be wrong.

A jig moves the accuracy problem off the part and into a fixture. The hole pattern is established once, in the jig, and then transferred identically to every part and every end. The bolt circle positions follow directly from the pattern geometry,

$$\theta_i = \frac{360^\circ}{n}\,i, \qquad x_i = R\cos\theta_i, \qquad y_i = R\sin\theta_i$$

where n is the number of fasteners and R the bolt circle radius. Locating off the jig also constrains the drill against wandering at entry, which on a curved surface is the most likely way to lose position.

Spending part of a very short night building a fixture rather than drilling immediately was the decision that mattered. It cost time up front and bought back accuracy and a second identical end.

Outcome

The vessel was drilled, the bulkheads fit on both ends, and the pressure vessel was ready by morning. We travelled on schedule and ran the hot fire.

What I would do differently

The jig should have existed before the night it was needed. The requirement was knowable weeks earlier. What this taught me is to look ahead at which operations on the critical path have no tooling yet, because those are the ones that turn into overnight problems.

Bulkhead holes located on the pressure vessel.

ST-01

Composite Overwrapped Pressure Vessel Study

Carbon fiber overwrap testing for higher tank pressure

ROLE	ORGANIZATION	PERIOD
General Member	Highlander Space Program, UC Riverside	2024 - 2025

The problem

Project Poseidon fed its engine from pressurized propellant tanks. Higher tank pressure raises chamber pressure, and higher chamber pressure raises thrust. What limits tank pressure is the tank wall itself: push a metal vessel too far and the wall stress reaches the material's yield point. You can allow more pressure by making the wall thicker, but that adds weight the vehicle then has to carry all the way up.

The question our small group set out to answer was whether a composite overwrapped pressure vessel, a COPV, could hold higher pressure without that weight penalty. I joined the effort as a general member.

Constraints

- **General-member role.** I was not leading this. I owned the CAD, and I shared the build and test with two teammates.
- **Student fabrication.** The test vessels, bulkheads, and tooling all had to be things we could machine and assemble ourselves.
- **A process we were still learning.** None of us were experienced with carbon fiber layup, so getting the process right was part of what we were testing rather than something we could assume.
- **Pressurized test article.** Anything that holds pressure is a hazard, so the parts had to be built and handled with that in mind.

Design decisions

Why a composite overwrap instead of a thicker metal wall

A thin-walled pressure vessel carries a hoop stress that grows with pressure and radius and falls with wall thickness,

$$\sigma_h = \frac{p\,r}{t}$$

To allow more pressure in an all-metal vessel you raise t, and that adds mass around the entire vessel. A COPV takes a different route. A thin metal liner holds the gas and provides the seal, while a carbon fiber overwrap carries most of the pressure load in tension. Carbon fiber has a very high strength for its weight, so the overwrap buys pressure capacity for far less mass than the equivalent metal would. That trade is the reason COPVs are used on flight vehicles in the first place.

Capping the vessel with bulkheads and a drill jig

To turn a plain section of tube into a vessel we could actually pressurize, both ends had to be closed. I designed two bulkheads in SolidWorks to cap the ends of a smaller test vessel, and a drill jig to locate the fastener holes.

The jig was the part that mattered most. The holes in the bulkhead have to line up with the holes in the vessel, and drilling them by hand without a guide would not hit the same pattern twice. Locating them with tooling instead of by eye is the same reasoning I used again later on the Bulkhead Drill Jig for Poseidon.

Building the overwrap by hand

With the vessel assembled, we wrapped carbon fiber around one of the two test vessels and left the other bare as a reference. The goal was to learn the process directly: how the fiber is tensioned and oriented, how the layup goes down, and how the overwrapped vessel behaves under pressure compared to the bare one.

What I built

Two bulkheads and a drill jig, all designed in SolidWorks, that turned a small tube into a sealed, testable pressure vessel. Working with two teammates, I helped assemble the vessels and overwrap one of them in carbon fiber to make the COPV test article.

Validation

We pressurized both articles and compared the bare vessel against the carbon fiber overwrapped one, so we could see directly how much pressure capacity the overwrap added rather than estimating it. That comparison, together with what it took to build and handle a COPV as a student team, is what the program used to weigh whether a COPV belonged on Poseidon.

What I would do differently

I would instrument the pressure test more heavily. We compared the two articles, but with more measurement across the wall we would have understood how the overwrap carried load, not only the point it reached, and we would have had data to design the next article against.

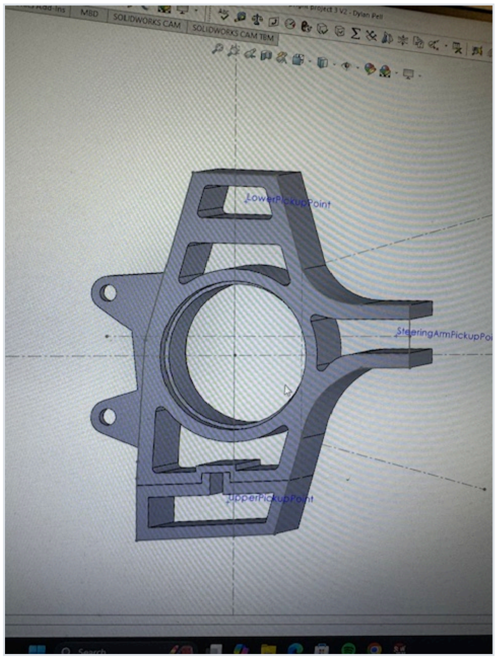
The more honest lesson is about what the project was really for. We did not end up putting a COPV on Poseidon. What we kept was the carbon fiber capability: we used what we learned about composite layup to build lightweight housings for the fluid components exposed underneath the pressure vessel, on the line running to the combustion chamber. The COPV was the goal we set, but the composites skill was the result we actually carried forward, and it went on to fly on Project Poseidon.

MD-01

Formula SAE Suspension Upright

Three design iterations for manufacturability

ROLE	ORGANIZATION	PERIOD
Suspension Engineer	Formula SAE, UC Riverside	2024 - 2025



The problem

The upright carries wheel loads into the suspension and locates the hub, so it needs tight tolerances on its critical features. The first design achieved that geometry but was expensive to make: complex features and awkward tool access meant more machining operations than the part warranted.

A part that cannot be produced accurately and repeatably with the equipment actually available is not a finished design, regardless of how it performs in CAD.

Design decisions

Across three iterations I worked the geometry against the manufacturing process rather than only against the load case. The levers that mattered:

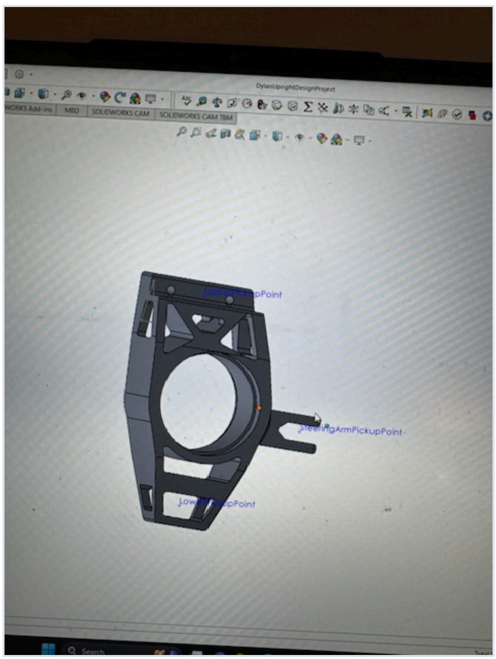
- **Reducing the number of setups.** Every time a part is unclamped and re-fixtured, cost goes up and a new tolerance stack is introduced between features cut in different orientations. Consolidating features into fewer orientations improves both cost and achievable accuracy.
- **Improving tool access.** Features that force long, thin tooling or unusual approach angles machine slowly and chatter. Opening up access let standard tooling reach the features.

- **Simplifying geometry that was not carrying load.** Complexity that existed for its own sake was removed; complexity that carried wheel loads stayed.

The net result was roughly a 25% improvement in manufacturability by reduced machining complexity and better feature accessibility, while preserving the precision required at the hub and mounting interfaces.

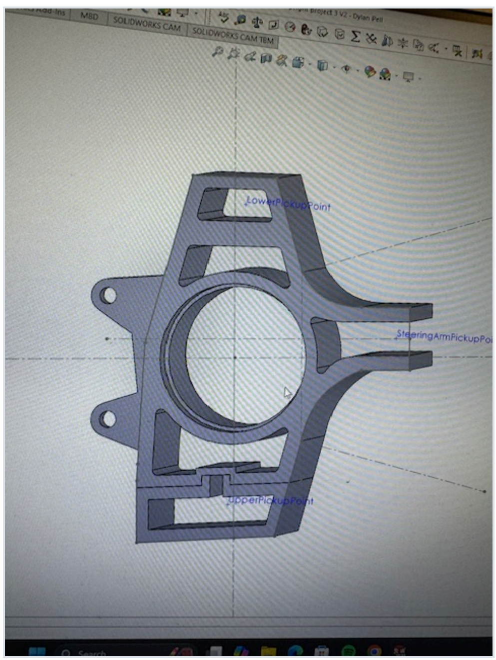
What I would do differently

I would involve whoever is cutting the part in the first iteration rather than the second. Most of what I found in cycles two and three, a machinist would have flagged in five minutes looking at cycle one. Design for manufacture works far better as a conversation than as a solo review.



First iteration in SolidWorks.

Second iteration.



Third and final iteration, the design I carried forward.